

SQL INDUSTRIAL PRACTICAL TASK

E-Commerce Product Inventory Management - DDL + DML + Operators

60 Small Practical Tasks - Basic to Intermediate

Student Name _____

Mobile Number _____

Batch _____ Date _____

Objective: Practice SQL DDL, DML, SELECT, filtering, operators, UPDATE, DELETE and ALTER TABLE operations using one simple e-commerce inventory-management database. No joins, GROUP BY, HAVING, subqueries or advanced reporting are required.

Suggested target for 1 hour: Minimum Task 1-40, Good Task 1-50, Excellent all 60.

Part A - Database & Table Creation

Task 1: Create a database named **ecommerce_db**.

Task 2: Use/select the **ecommerce_db** database.

Task 3: Create a table named **products** with the following columns:

```
product_id INT PRIMARY KEY
product_name VARCHAR(100)
category VARCHAR(50)
brand VARCHAR(50)
price DECIMAL(10,2)
quantity INT
city VARCHAR(50)
status VARCHAR(20)
```

Task 4: Display the structure of the **products** table.

Task 5: Display all tables available inside **ecommerce_db**.

Part B - INSERT Operations

Task 6: Insert the following product record:

201, Galaxy M55, Mobile, Samsung, 32000, 15, Pune, Available

Task 7: Insert the following product record:

202, iPhone 15, Mobile, Apple, 65000, 8, Mumbai, Available

Task 8: Insert the following product record:

203, Moto Edge 50, Mobile, Motorola, 28000, 20, Pune, Available

Task 9: Insert the following product record:

204, Inspiron 15, Laptop, Dell, 58000, 6, Nashik, Available

Task 10: Insert the following product record:

205, IdeaPad Slim 3, Laptop, Lenovo, 45000, 12, Mumbai, Available

Task 11: Insert the following product record:

206, Galaxy Watch 6, Watch, Samsung, 22000, 4, Pune, Out of Stock

Part B - INSERT Operations (Continued)

Task 12: Insert the following product record:

207, Apple Watch SE, Watch, Apple, 30000, 10, Mumbai, Available

Task 13: Insert the following product record:

208, Redmi Pad, Tablet, Xiaomi, 24000, 18, Pune, Available

Task 14: Insert the following product record:

209, OnePlus Pad, Tablet, OnePlus, 35000, 5, Nashik, Available

Task 15: Insert the following product record:

210, Bluetooth Speaker, Accessories, JBL, 7000, 25, Mumbai, Available

Part C - Basic SELECT & Operator Practice

Task 16: Display all records from **products**.

Task 17: Display only **product_name**.

Task 18: Display **product_name** and **price**.

Task 19: Display **product_name**, **category**, **brand** and **price**.

Task 20: Display products available in Pune.

Task 21: Display products available in Mumbai.

Task 22: Display all Mobile products.

Task 23: Display all Laptop products.

Task 24: Display products whose price is greater than 30000.

Task 25: Display products whose price is less than 30000.

Task 26: Display products whose price is equal to 35000.

Task 27: Display products whose price is greater than or equal to 45000.

Task 28: Display products whose price is less than or equal to 30000.

Task 29: Display products whose quantity is greater than 10.

Task 30: Display products whose quantity is less than 10.

Part D - Logical Operators

Task 31: Display products from Pune **AND** Mobile category.

Task 32: Display products from Mumbai **AND** Available status.

Task 33: Display products where **price > 30000 AND quantity > 5**.

Task 34: Display products where **price >= 30000 AND price <= 60000**.

Task 35: Display products from Pune **OR** Mumbai.

Part D - Logical Operators (Continued)

Task 36: Display products belonging to Mobile **OR** Laptop category.

Task 37: Display products where **quantity < 10 OR price > 50000**.

Task 38: Display products where **category = Mobile AND price > 30000**.

Task 39: Display products where **brand = Samsung OR brand = Apple**.

Task 40: Display products where **city = Pune AND status = Available AND quantity > 10**.

Part E - Slightly Higher Operator Logic

Task 41: Display products whose price is **BETWEEN 25000 and 50000**.

Task 42: Display products whose quantity is **BETWEEN 5 and 15**.

Task 43: Display products whose category is **IN** Mobile, Laptop, Tablet.

Task 44: Display products whose city is **IN** Pune, Mumbai.

Task 45: Display products whose brand is **NOT** Samsung.

Task 46: Display products whose status is **NOT** Out of Stock.

Task 47: Display products whose price is **NOT EQUAL TO** 30000.

Task 48: Display products whose product name **starts with** Galaxy.

Task 49: Display products whose product name **contains** Pad.

Task 50: Display products where **category = Mobile AND (price > 30000 OR quantity > 15)**.

Part F - UPDATE Operations

Task 51: Update the price of Galaxy M55 from 32000 to 34000.

Task 52: Update the quantity of iPhone 15 to 12.

Task 53: Change the status of Galaxy Watch 6 from Out of Stock to Available.

Task 54: Increase the price of all Mobile products by 2000.

Task 55: Increase the quantity of all Pune products by 5.

Task 56: Update the status to Out of Stock for products whose quantity is less than 5.

Part G - DELETE Operations

Task 57: Delete the product whose product_id = 210.

Task 58: Delete products whose price is less than 8000.

Task 59: Delete products where status = Out of Stock AND quantity < 5.

Task 60: Delete products belonging to the Tablet category whose price is greater than 30000.

Topics Used in This Practical

```
CREATE DATABASE
USE
CREATE TABLE
DESC / DESCRIBE
SHOW TABLES
INSERT
SELECT
WHERE
AND
OR
NOT
BETWEEN
IN
```

LIKE
UPDATE
DELETE
ALTER TABLE
ADD COLUMN
MODIFY COLUMN
RENAME COLUMN
DROP COLUMN

Reminder: Write the SQL query yourself for every task. Do not use JOIN, GROUP BY, HAVING, subqueries, stored procedures, views or advanced functions.